

(a) textbook ABC-PMC, schematic: one update per generation shown: generation $t \rightarrow t+1$, after 135 simulations

P_{t+1} : the N kept, weighted; the rejected are discarded

barrier: q_{t+1} needs the complete P_{t+1}

$$\epsilon_{t+1} = \text{quantile}_{\alpha} \{\rho_i\}_{i \leq N}$$

hard: keep $\rho < \epsilon_{t+1}$

$$q_t = \sum_{i \leq N} \omega_i^{(t)} K_{\Sigma_t}(\cdot - \theta_i)$$

$$\omega^{(t+1)} \propto \pi/q_t$$

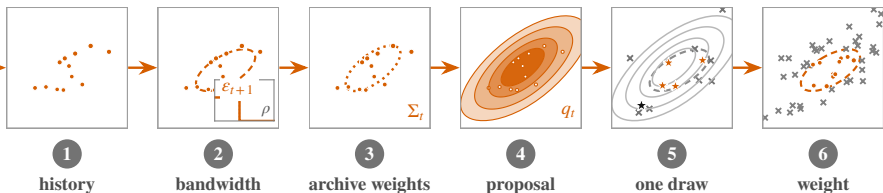
after acceptance

$P_t: (\theta_i, \rho_i, \omega_i^{(t)})$
gen. $< t$ discarded

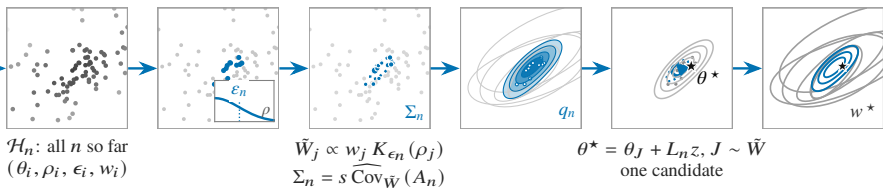
$$\Sigma_t = 2 \widehat{\text{Cov}}_{\omega^{(t)}}(\{\theta_i\})$$

weights carried over

$\theta^* \sim q_t$, simulate,
keep if $\rho^* < \epsilon_{t+1}$:
one at a time, until N kept



(b) generation-free (ours): one update per arrival shown: the update after 84 evaluations



$$A_n = \text{Top}_k(\rho)$$

ϵ_n non-increasing, smooth K_{ϵ_n}

$$q_n = \sum_{j \in A_n} \tilde{W}_j K_{\Sigma_n}(\cdot - \theta_j)$$

$$w^* = \pi/\bar{q}_n^{\text{on}} \text{ at } \theta^*$$

θ^* to a free worker; ρ^* arrives later, among other workers' results: append $(\theta^*, \rho^*, \epsilon_n, w^*)$

bootstrap, fewer than k particles: prior draw, $w^* = 1$, no bandwidth stored

every panel: the same unit prior box